airtel

SYNC

ENTERPRISE TRACK A · PROJECT UDYAM DIGITAL

Udyam Digital

A small business goes live in one sitting: on a phone, in Hindi or English, with no salesperson in the loop.



Live prototype udyam-digital-isyfatadrab4adtbqddzwp.streamlit.app
Code github.com/Sheel23501/udyam-digital

Team: Sheel Gautam (IIIT Delhi) and Vikas Meena · November 2026

Platform Partner unstop

₹35.6 lakh of ₹55.4 lakh went to leads that never activated

4,632 of 5,000 leads dropped last quarter. Half of those drops were the sales process itself, not the offer.

48%

of drops: unreachable, no time for a visit, visit not scheduled, quote too slow or too complex

₹53,026

field-sales cost per activation on micro retail, for a ₹2,619/month customer

17 days

median lead-to-activation; the quote alone takes 4 days

719 → 9

owners asked for 5G AirFiber 719 times; sales sold it 9 times

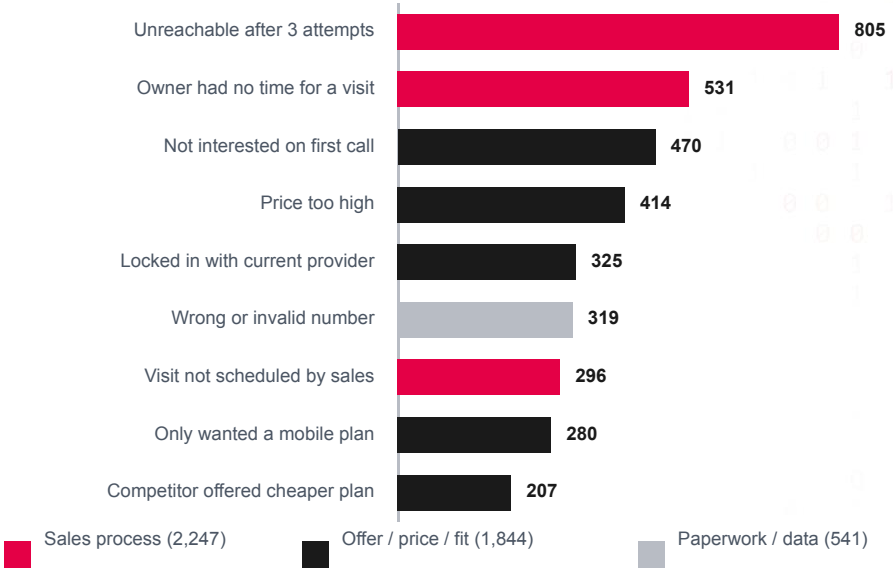
22

customers activated last quarter with a cancelled or suspended GSTIN

6% → 28%

6-month churn as the status calls a customer had to make rose from 1 to 5+

Why 4,632 leads dropped (top 9 reasons)



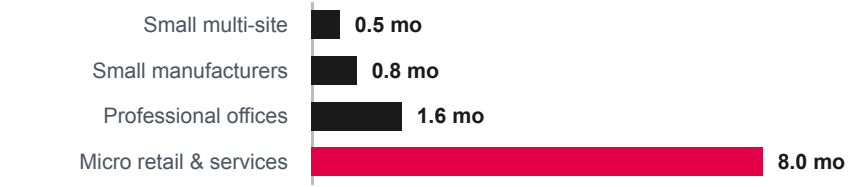
Source: smb_leads.csv, 5,000 leads (2026-04-01 to 2026-06-30); every figure reproduced by backend/scripts/make_charts.py.

Where to go first: payback decides

Blended CAC per activation (all leads' acquisition cost ÷ activations) against the monthly bundle value, by segment.

Segment	Leads	Activation	₹/month	6-mo churn	CAC / activation	Payback	Our move
Small multi-site	755 (15%)	11.0%	₹31,233	11%	₹15,614	0.5 mo	Wave 1: digital discovery, field close
Small manufacturers	289 (6%)	12.8%	₹19,980	8%	₹15,324	0.8 mo	Wave 2: field-assisted (ILL, SD-WAN)
Professional offices	1,247 (25%)	11.1%	₹6,796	14%	₹10,695	1.6 mo	Wave 1: digital, optional assist
Micro retail & services	2,709 (54%)	4.1%	₹2,505	23%	₹20,007 (field ₹53,026)	8.0 mo	Digital only; no field visits

Months of revenue to repay the acquisition cost



Verdict. Professional offices and small multi-site first: they pay back in weeks. Micro retail goes digital-only, where a field visit costs ₹53,026 to win ₹2,619 a month. Manufacturers come in wave 2 with field assistance for leased lines and SD-WAN.

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One bundle per segment, one monthly number

Composed by the rules engine from the catalogue's eligibility rules and minimum prices; the owner can switch items on and off and the price re-validates instantly.

Dukaan Start

Micro retail · GST-free path

from **₹1,198** /month

Setup ₹0 · 6-month · live in 2 days

- Biz 5G AirFiber 100
- Business Postpaid 299

Live in 2 days, no setup fee, no GST needed. For the 1,466 no-GST leads (2.0% activation today).

Clinic / Office Pro

Professional offices

from **₹5,487** /month

Setup ₹1,500 · 12-month · live in 5 days

- Biz Fibre 300
- Business Postpaid 599 ×2
- Cloud Email & Office Suite ×8
- Cloud Backup 1 TB
- Endpoint Security ×8
- Backup 4G/5G Failover

Email for every user, backup, endpoint security and a failover line, because a clinic cannot be offline.

Multi-Store

Small multi-site · 3 outlets

from **₹21,579** /month

Setup ₹24,800 · 24-month · live in 14 days

- Biz Fibre 300 ×3
- Business Postpaid 599 ×6
- Backup 4G/5G Failover ×3
- Managed SD-WAN ×3
- Toll-free Number
- Cloud CCTV ×6
- Cloud POS & Billing ×3

Every outlet on one network with cameras and billing that never stops; SD-WAN and toll-free need GST.

Factory Connect

Small manufacturers

from **₹14,377** /month

Setup ₹17,000 · 24-month · live in 21 days

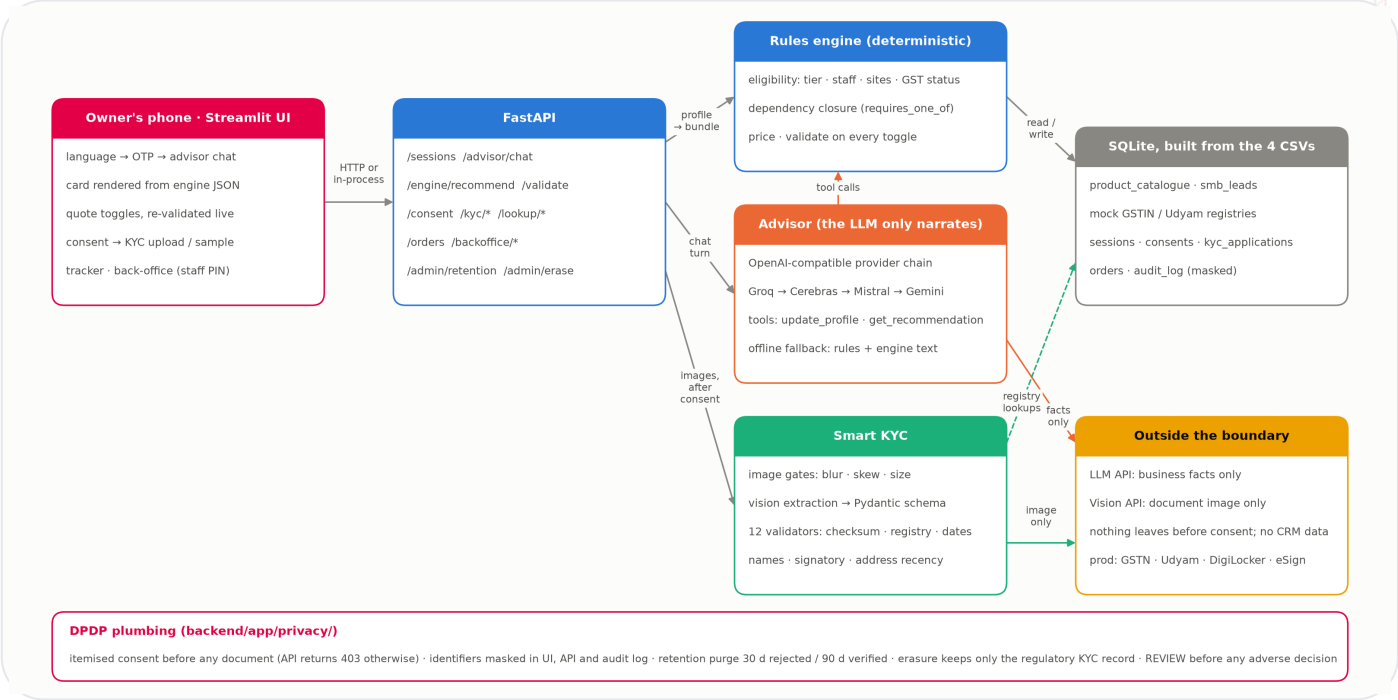
- Internet Leased Line 20
- Business Postpaid 599 ×3
- Machine Monitoring IoT ×6
- Endpoint Security ×12
- Cloud Backup 1 TB
- Managed Firewall

A leased line with an SLA, machine monitoring and a managed firewall; 21-day install stated up front.

Prices are catalogue minimums composed by backend/app/engine (python scripts/run_scenarios.py reproduces them); ranges and setup fees shown to the owner on the quote screen.

How it is built: rules decide, the LLM narrates

1. **Enter**
   link, QR or WhatsApp; mobile + one-time code
2. **Advisor**
   4 to 6 questions in Hindi or English
3. **Quote**
   one monthly number, re-checked on every toggle
4. **Consent**
   itemised DPDP notice before any document
5. **Smart KYC**
   four photos, 12 checks in seconds
6. **Tracker**
   dated stages, zero status calls



Consent gates every document; the engine still quotes if the model is down. Full-resolution diagram: docs/architecture.png (submitted separately).

Smart KYC: 12 checks in seconds, built around the DPDP Act

Vision extraction of four documents, then deterministic validators against the GST and Udyam registries. A photo problem is a REVIEW, never a rejection.

The 12 sample applicants as the prototype grades them: 12/12 match the key

FAIL **K01** GSTIN cancelled; Udyam number missing	REVIEW **K02** GST certificate photo rotated	PASS **K03** all documents consistent
REVIEW **K04** GST certificate photo blurred	FAIL **K05** Udyam certificate missing	FAIL **K06** GSTIN check digit fails: tampered digit
FAIL **K07** legal name differs across documents	PASS **K08** all documents consistent	PASS **K09** all documents consistent
FAIL **K10** signatory is not the person of record	FAIL **K11** electricity bill 257 days old (limit 90)	PASS **K12** all documents consistent

22 → 0

activations with a cancelled or suspended GSTIN: the registry is checked live, before the order

DPDP controls, each a code path, not a slide

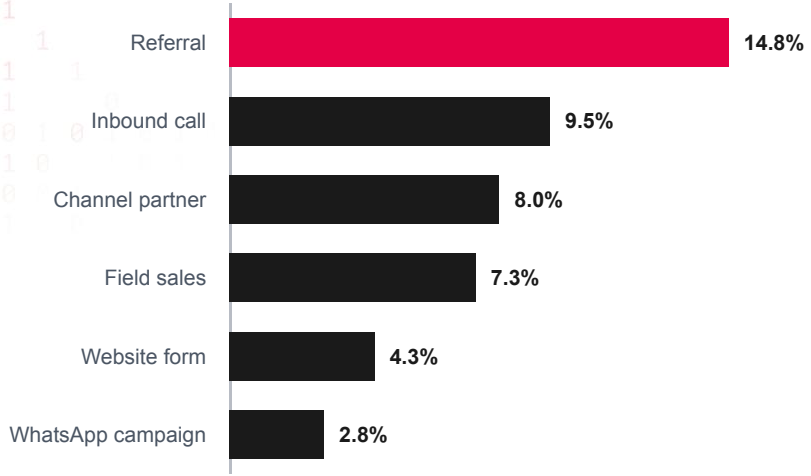
Consent gate: documents are refused (HTTP 403) until the owner ticks four itemised purposes in their language
Minimisation: five facts for the quote; GST or Udyam path; no Aadhaar; only the image leaves the boundary
Masking: GSTIN as 27XXXXXXXXXX1ZP, names as 'Nisha A.' in UI, API and audit log
Retention: images deleted 30 days after a rejection, 90 after verification; erasure on request
Accuracy: REVIEW before any adverse decision; 'talk to a human' always available

Reproduce: cd backend && python scripts/kyc_report.py --key tests/kyc_answer_key_images.json. Privacy note: docs/privacy-note.md (submitted separately).

Digital adds; the field force multiplies

Take the field team out of the ₹2,500 kirana sale and put it where payback is under a month. Attribution keeps partners on side.

Activation rate by lead source



Referral converts at 14.8% but churns at 24.6%: pay the referral bonus at month 3, not at activation.

Micro retail goes digital-only

Field sales spent ₹53,026 per activation here. Removing those 27 visits a quarter saves about ₹14 lakh per quarter with no loss of volume.

Field team redeployed to multi-site and manufacturers

Payback 0.5 and 0.8 months. The advisor qualifies and quotes; a human closes the ₹20,000-plus accounts and handles ILL and SD-WAN installs.

Partners keep their code

Every journey carries a partner code for attribution and commission; partners see the scored queue for their area.

Nothing dead-ends

Leads the journey cannot close land in a back-office queue ranked by a propensity model (holdout AUC 0.68) plus how recoverable the drop reason is, each with a suggested next action.

Source: smb_leads.csv, 5,000 leads (2026-04-01 to 2026-06-30); every figure reproduced by backend/scripts/make_charts.py.

Rollout in three phases

Start where the data says the money is, with the adapters that make mocks real.

Phase 0 · Q3 2026

Prove it

- Delhi + Mumbai
- Hindi and English, Hinglish auto-detect
- Micro retail (GST-free) + professional offices
- Web and WhatsApp entry, partner-code pilot with 20 partners
- Mock registries swapped for GSTN/GSP and Udyam APIs

Gate: 12% blended activation, quote in under 5 minutes, zero activations with a non-active GSTIN

Phase 1 · Q4 2026

Widen

- Tier-2 cities (Jaipur, Lucknow, Indore, Nagpur, Surat...)
- Small multi-site journeys with per-site quotes
- DigiLocker pull of GST/Udyam certificates, eSign for the order
- Field team re-routed by the propensity queue

Gate: CAC per activation ≤ ₹7,500; KYC first-pass ≥ 70%

Phase 2 · Q1 2027

Deepen

- Manufacturers: leased line and SD-WAN journeys with field assist
- Real-time fibre serviceability at quote time
- 30-day check-in and referral bonus at month 3
- Regional languages beyond Hindi where volume justifies it

Gate: 6-month churn ≤ 10%; status calls per customer < 0.5

Dependencies: GSTN/GSP and Udyam API access, DigiLocker and eSign onboarding, a DLT-registered SMS template for OTPs, partner commission rules.

What we will measure

Q2 2026 baseline from the leads file; two-quarter targets, and the design feature that moves each one.

Metric	Today	Target (2 quarters)	What moves it
Lead → activation	7.4%	12% blended; 15% for offices and multi-site	Instant quote, GST-free path, no visit-dependent steps
Lead → activation time	17 days (quote alone 4)	≤ 5 days, install-bound; quote in < 5 minutes	Advisor + engine + instant KYC
Status calls per customer	3.0	< 0.5	Tracker with dated stages and proactive notifications
Blended CAC per activation	₹15,053	≤ ₹7,500 (micro ≤ ₹3,000)	Digital-only micro, referral, redeployed field force
6-month churn	15.2%	≤ 10%	Fewer calls, right-sized bundles, 30-day check-in, bonus at month 3
Activations with non-active GSTIN	22	0	Live registry check inside KYC, before the order
KYC first-pass rate	n/a	≥ 70%; REVIEW resolved within 24 h	Human-language fixes, re-upload path, Udyam-only path

Measured weekly from the same tables the prototype already writes: sessions, consents, kyc_applications, orders, audit_log.

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Risks we designed for, and what is already built

Each mitigation is a code path in the prototype, not a plan.

Risk	Mitigation in the prototype
The model recommends something the owner is not eligible for	The rules engine decides every item; the model only narrates; the UI renders engine JSON. 30 scenario fixtures + 16 live transcripts, zero violations
KYC wrongly rejects a genuine owner	Photo problems are REVIEW with a plain-language fix; FAIL always carries the reason and a 'talk to a human' hand-off
LLM or vision API down or over quota during a demo	Provider chain with fallbacks; cached demo replies; the engine and validators run offline. The journey completes with no model at all
Fibre not feasible, install delays	5G AirFiber as a 2-day interim line; install days stated on the quote; tracker dates
Channel conflict with partners and field sales	Partner code on every journey; propensity queue routes high-value leads to humans; micro retail digital-only
Privacy and DPDP exposure	Consent gate (403), masking everywhere, retention purge, erasure, staff-only back office, audit log
Low-quality leads from digital channels	Mobile number verified by one-time code at entry; GST status checked live

Built and verified

2,050 automated tests, all green in CI

30 / 30 advisor scenarios through the rules engine

12 / 12 KYC verdicts matching the answer key

16 / 16 live conversations on Gemini, zero rule violations

Docker compose build and smoke test on every push

Try it: the README's step-by-step guide takes a judge from clone to a verified API in 20 minutes, no keys needed.

Repository: github.com/Sheel23501/udyam-digital · docs/how-to-check.md · docs/privacy-note.md · docs/architecture.png